

Parvinder Singh

Richardson, Texas • parvinder.s3468@gmail.com • +1 (469) 465-1019 • LinkedIn

Summary

Data Engineer & Analyst with around 4 years of experience in building scalable ETL pipelines, optimizing databases, and delivering actionable insights. Proficient in Python, SQL, and AWS tools (Glue, Redshift, S3) for data processing and warehousing. Skilled in creating Tableau dashboards and performing statistical analysis to drive business decisions. Experienced in machine learning workflows (scikit-learn, XGBoost) and cloud platforms (AWS, Snowflake). Combines technical expertise with strong collaboration skills to solve complex data challenges and deliver measurable results.

Education

Master of Science in Computer Science University of Texas at Dallas, Richardson, Texas, USA	Aug 2021 - Dec 2023
Bachelor of Engineering in Computer Engineering University of Pune, Maharashtra, India	Aug 2016 - May 2020

Skills

- **Programming Languages:** Python, SQL, R
- **Tools & Collaboration:** Git, Jira, Jenkins, Excel (Power Pivot, VBA, Advanced), Docker
- **Data Visualization & BI Tools:** Tableau, Power BI, Matplotlib, Seaborn, Looker, QlikView
- **Databases & Warehousing:** MySQL, Google Big Query, Amazon Redshift, PostgreSQL, Oracle
- **Machine Learning:** Scikit-learn, XGBoost, Logistic Regression, Random Forest, Model Evaluation
- **Cloud Platforms:** AWS (S3, Lambda, Redshift), Snowflake, Azure (Synapse, Data Factory), Databricks
- **Data Analysis & Statistics:** Pandas, NumPy, Statistical Analysis, A/B Testing, SAS, SPSS, Predictive Modelling
- **ETL & Data Engineering:** AWS Glue, PySpark, Apache Airflow, Azure Data Factory, Batch/Stream Processing

Experience

Northern Trust, Tx, USA Data Engineer	Feb 2024 - Current
• Collaborated with engineering teams to design MySQL database schemas for customer review data storage, optimizing OLTP performance through indexing and partitioning, reducing query latency by 18%. • Built ETL pipelines using SQL to aggregate and transform raw data from multiple tables, enabling seamless integration into data lakes (AWS S3) and improving pipeline reliability with data validation checks. • Engineered batch processing workflows (Python, Hadoop) to ingest and clean structured/unstructured data from MySQL and MongoDB, resolving schema mismatches and ensuring data consistency for downstream analytics. • Supported BI initiatives by analysing marketing campaign metrics (e.g., CTR, conversion rates) using R, automating weekly reports that reduced manual effort by 10 hours/month. • Developed Tableau dashboards to visualize customer sentiment trends, enabling stakeholders to identify actionable insights and driving a 15% improvement in campaign ROI. • Implemented feature engineering pipelines (Python) for ML models, including normalization, tokenization, and PCA-driven dimensionality reduction, improving model training efficiency by 25%. • Contributed to ML model development (scikit-learn) by training and validating Logistic Regression and Random Forest classifiers, achieving 88% accuracy in customer churn prediction. • Built and deployed Spark-based ETL workflows using PySpark, streamlining the migration of medical data from legacy systems to AWS One Lake, improving data processing efficiency by 40%. • Orchestrated data workflows using Airflow to automate ETL jobs, reducing manual intervention and ensuring SLA-compliant daily data refreshes for reporting.	
Hexaware Technologies, India Data Analyst	Sep 2018 - Jul 2021
• Migrated and cleaned legacy datasets using Python (Pandas) and SQL, resolving data inconsistencies to improve data integrity; reduced manual errors in ETL workflows by 15% through automated validation checks. • Supported dimensional modelling for a sales analytics warehouse in SQL Server, designing star-schema tables that improved query efficiency by 12% and aligned with Kimball methodology. • Automated CSV-to-warehouse data ingestion using Python scripts, reducing manual processing time by 20% and enabling scheduled daily updates for finance reporting. • Created Tableau dashboards to visualize customer retention KPIs, reducing ad-hoc report requests by 30% and enabling stakeholders to self-serve insights. • Conducted data quality checks on CRM datasets using SQL, identifying anomalies (e.g., mismatched customer IDs) and improving alignment with business logic for targeted campaigns. • Integrated Salesforce APIs with internal databases to enrich customer profiles, reducing manual data entry by 10 hours/month and enabling automated weekly syncs. • Performed exploratory data analysis (EDA) on survey data using Excel and Power BI, transforming unstructured inputs into actionable insights for product teams. • Developed and optimized large-scale data pipelines on AWS, enhancing the ingestion and transformation of clinical data such as EHR and claims data, reducing ETL execution time by 30% using S3 and EMR for efficient data processing. • Improved query performance on relational and NoSQL databases like MySQL and MongoDB, increasing data retrieval speed by 25%, while automating data processing with AWS Lambda for faster healthcare insights. • Monitored and fine-tuned Spark and Hadoop jobs using AWS CloudWatch, optimizing resource allocation and cutting infrastructure costs by 20%, ensuring seamless data flow for clinical decision support systems.	